

Microprocessor System Lab8 Check List

- **Date:** 2024/11/19
 - **Please follow the instructions, and report to TAs after finishing check points.**
 - **If you have any questions or troubles during the experiment, please feel free to ask TAs or the teacher, discussion with classmates is also encouraged.**
 - **Please refer to the content from class slides, you can also use the resources from the internet.**
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Part I. Question

Please refer to the class content and answer the following questions.

Check Points:

- 1.1:** Please try to explain what is the difference between level-triggered and edge-triggered?

Part II. Lab8 Experiment

This lab is external interrupt & PWM, please download the source codes on the google cloud and check the result after execution.

Check Points:

- 2.1:** Please run the “External Interrupt” in the example project. Observe the two function of external interrupt.

Hint: Difference between level-trigger and edge-trigger.

- 2.2:** Please run the three projects in “PWM” in the example project. Observe the difference between the three kinds of LED behavior.

Hint: We can change the LED luminance by tuning duty cycle.

Part III. Practice

Please reference the sessions and source code of Part II. Try to design and implement the Practice. (You can check the executed result from google cloud)

- 3.1:** Please use timer and external interrupt to realize LED PWM. Use one edge-triggered external interrupt as start/pause of LED rotating control and one level-triggered external interrupt to light up all LEDs. For LED rotating control, there are 4 modes, the following are details about those modes.

First mode is LED dimming at original location. Second mode is LED dimming left-shift, and when it comes to the left-most position, rotate to the right-most position. Third mode is LED dimming right-shift, and when it comes to the right-most position, rotate to the left-most position. Forth mode is like ping-pong, that is right-shift to the right-most position, and start to left-shift to the left-most position, and start to right-shift, and so on. You can refer to demo video to better understanding.